COMPLEX DIFFERENTIAL GEOMETRY ON THE HELICAL MANIFOLD
PARTS I–IV: SINGLE CELL & STACKED CHAIN — AXIOMS 1–6, COROLLARIES 1–20
CONTRACTION INTEGRALS THROUGHOUT · LEIBNIZ NOTATION ONLY

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Prerequisite Reading

This paper uses contraction integrals exclusively—no Riemann sums appear anywhere. The contraction integral is an alternative geometric framework for integration based on level-set contraction and the layer-cake decomposition. Its central tool is inverse duality: for a strictly monotone continuous f on $[a, b]$,

$$\int_a^b f + \int_{f(a)}^{f(b)} f^{-1} = b \cdot f(b) - a \cdot f(a).$$

The companion paper, "The Contraction Integral: Integration via Level-Set Contraction" (Yazdani, 2026), develops this framework from first principles and should be read first.

Abstract. Six axioms define a helical 2-manifold embedded as a surface in $\mathbb{R}^3$. From these axioms, together with standard differential geometry (explicitly imported where required), we obtain: an induced metric with $g^{uu} = \frac{1}{2}$ (fixing $c = 1$); a constant gauge connection $A_u = 2\pi k$ whose holonomy identifies k as a topological winding number; a gauge-diagonalised cone metric; geodesic-derived radial dynamics with an effective potential $V(u) = k^2/(2(1+u)^2)$; a proof that local physics is cell-independent; integer-quantised angular momentum from the closure condition; a per-mode radial ODE (Euler's equation on the cone); and a phase-matching condition $\ell = k$ from holonomy coherence.

The radial ODE admits: interpretation as the separated form of a wave equation on the manifold; a scalar first-order factorisation analogous to relativistic factorisation procedures; and reduction to a Schrödinger-type normal form.

The Laplace–Beltrami operator and wave equation $\Box_M \psi = 0$ are imported from standard Riemannian geometry and are not derived from the axioms. Any physical interpretation (Klein–Gordon, Dirac, Schrödinger analogies) is presented as structural correspondence, not equivalence.

All integrals are contraction integrals computed via inverse duality. All derivatives in Leibniz notation. One manifold. One parameter.

1. The Six Axioms

Axiom 1 (Time is Self-Referential)**.** $\dfrac{dt}{dt} = 1.$

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This fixes $c = 1$ in natural units. The z-axis grows as $z = 1 + u$, with $u \in [0, 1]$ the affine parameter within one cell.

Axiom 2 (Coupled Radius)**.** *Radius is derived from the angular variable ($r = \theta/\pi$) and the spiral expands as a light cone: $dr/dz = 1$. For one cell:*

$$r(u) = 1 + u.$$

Axiom 3 (Light-Cone Expansion)**.** *The radial expansion rate equals the speed of light: $dr/dz = 1$. Combined with Axiom 2, this couples the transverse radius to the longitudinal coordinate.*

Axiom 4 (Helical Motion)**.**

$$\theta(u, v) = v + 2\pi k\, u, \qquad k \in \mathbb{Z}^+.$$

The winding number k is the single free geometric parameter.

Axiom 5 (Reality Embedding)**.**

$$\mathbf{R}(u, v) = \big((1 + u)\cos\theta,\ (1 + u)\sin\theta,\ u\big), \qquad \theta = v + 2\pi k\, u.$$

Axiom 6 (Unitary Closure)**.** *The spiral closes at integer z:*

$$\psi(u, v + 2\pi) = \psi(u, v).$$

This forces every transverse mode to carry integer angular momentum.

2. Corollaries

Corollary 1 (Induced Metric)**.** *The first fundamental form on M is computed by taking partial derivatives of the embedding $\mathbf{R}(u, v)$ with respect to u and v, then forming dot products.*

The tangent vectors are $\mathbf{R}_u = \partial\mathbf{R}/\partial u$, $\mathbf{R}_v = \partial\mathbf{R}/\partial v$.

The metric components $E = \mathbf{R}_u \cdot \mathbf{R}_u$, $F = \mathbf{R}_u \cdot \mathbf{R}_v$, $G = \mathbf{R}_v \cdot \mathbf{R}_v$ evaluate to:

$$E = 4\pi^2 k^2 (1 + u)^2 + 2, \tag{1}$$

$$F = 2\pi k\,(1 + u)^2, \tag{2}$$

$$G = (1 + u)^2. \tag{3}$$

The metric determinant and its square root:

$$\det(g) = EG - F^2 = 2(1 + u)^2, \qquad \sqrt{g} = \sqrt{2}\,(1 + u).$$

The inverse metric component:

$$g^{uu} = \frac{G}{\det(g)} = \frac{(1 + u)^2}{2(1 + u)^2} = \frac{1}{2}.$$

Derivation. The embedding gives:

$$\mathbf{R}_u = \big(\cos\theta - (1 + u)(2\pi k)\sin\theta,\ \sin\theta + (1 + u)(2\pi k)\cos\theta,\ 1\big),$$

$$\mathbf{R}_v = \big(-(1 + u)\sin\theta,\ (1 + u)\cos\theta,\ 0\big).$$

Computing dot products and applying $\sin^2\theta + \cos^2\theta = 1$*:*

$$\begin{aligned} E &= \cos^2\theta + (1+u)^2(2\pi k)^2\sin^2\theta - 2(1+u)(2\pi k)\cos\theta\sin\theta \\ &\quad + \sin^2\theta + (1+u)^2(2\pi k)^2\cos^2\theta + 2(1+u)(2\pi k)\sin\theta\cos\theta + 1 \\ &= 1 + 4\pi^2k^2(1+u)^2 + 1 = 4\pi^2k^2(1+u)^2 + 2. \end{aligned}$$

The cross terms in $\sin\theta\cos\theta$ *cancel. Similarly, the cross terms cancel in* F *and* G*, yielding the stated results. The determinant follows from direct computation:*

$$EG - F^2 = \big(4\pi^2k^2(1+u)^2 + 2\big)(1+u)^2 - 4\pi^2k^2(1+u)^4 = 2(1+u)^2.$$

Verified symbolically via SymPy.

Corollary 2 (Holonomy — Winding Number as Topological Charge)**.** *The off-diagonal metric component divided by the transverse metric component defines a gauge connection for the* v*-fibre:*

$$A_u = \frac{F}{G} = \frac{2\pi k\,(1+u)^2}{(1+u)^2} = 2\pi k.$$

Interpretation. A_u *measures how much phase (the* v*-direction) gets dragged per step in* u*. Because both* F *and* G *scale as* $(1+u)^2$*, their ratio is constant: the helix winds at a uniform rate regardless of where you are in the cell.*

The holonomy over one cell:

$$\Phi_{\text{cell}} = \int_0^1 A_u\,du = \int_0^1 2\pi k\,du = 2\pi k.$$

This is exact: k *complete phase cycles per cell. Since* $k \in \mathbb{Z}^+$*, the phase returns to itself (*$e^{i\cdot 2\pi k} = 1$*), making* k *a topological invariant—it cannot be changed by continuous deformation of the manifold.*

The holonomy selects k *as the fundamental geometric quantum number of the helix.*

Verified symbolically via SymPy.

Corollary 3 (Christoffel Symbols)**.** *Computed from the standard formula*

$$\Gamma^i_{jl} = \frac{1}{2}\sum_m g^{im}\left(\frac{\partial g_{mj}}{\partial x^l} + \frac{\partial g_{ml}}{\partial x^j} - \frac{\partial g_{jl}}{\partial x^m}\right),$$

using the metric (**??**)–(**??**) *and its inverse. The six independent symbols are:*

$$\Gamma^u_{uu} = -2\pi^2k^2(u+1), \qquad \Gamma^v_{uu} = \frac{4\pi k\big(\pi^2k^2u^2 + 2\pi^2k^2u + \pi^2k^2 + 1\big)}{u+1}, \tag{4}$$

$$\Gamma^u_{uv} = -\pi k\,(u+1), \qquad \Gamma^v_{uv} = \frac{2\pi^2k^2u^2 + 4\pi^2k^2u + 2\pi^2k^2 + 1}{u+1}, \tag{5}$$

$$\Gamma^u_{vv} = -\frac{u+1}{2}, \qquad \Gamma^v_{vv} = \pi k\,(u+1). \tag{6}$$

Verified symbolically via SymPy and cross-checked numerically at $u = 1/2$*,* $k = 1$*.*

Corollary 4 (Cell Action via Contraction)**.** *The action (surface area) of one cell:*

$$S = \int_0^{2\pi}\int_0^1 \sqrt{g}\,du\,dv = 2\pi\int_0^1 \sqrt{2}\,(1+u)\,du.$$

Set $h(u) = \sqrt{2}\,(1+u)$*. Then* $h(0) = \sqrt{2}$*,* $h(1) = 2\sqrt{2}$*, and* $h^{-1}(t) = t/\sqrt{2} - 1$*.*

By inverse duality:

$$\int_0^1 h(u)\,du = 1\cdot h(1) - 0\cdot h(0) - \int_{\sqrt{2}}^{2\sqrt{2}} h^{-1}(t)\,dt = 2\sqrt{2} - \frac{\sqrt{2}}{2} = \frac{3\sqrt{2}}{2}.$$

Therefore:

$$S = 2\pi \times \frac{3\sqrt{2}}{2} = 3\sqrt{2}\,\pi.$$

The action is independent of k. The winding number does not affect the surface area of the cell.

Verified symbolically via SymPy.

Corollary 5 (Phase Energy via Geodesic Derivation)**.** *Origin of the potential. The coordinate change $v' = v + 2\pi k u$ removes the off-diagonal metric term, yielding the cone metric:*

$$ds^2 = 2\,du^2 + (1+u)^2\,dv'^2.$$

The geodesic equations on this cone are:

$$\frac{d^2u}{ds^2} - \frac{(1+u)}{2}\left(\frac{dv'}{ds}\right)^2 = 0, \qquad \frac{d}{ds}\left[(1+u)^2\frac{dv'}{ds}\right] = 0.$$

The second equation gives conservation of angular momentum: $(1+u)^2\,dv'/ds = L = const.$ Substituting $dv'/ds = L/(1+u)^2$ into the first equation:

$$\frac{d^2u}{ds^2} = \frac{L^2}{2(1+u)^3}.$$

Reducing to first order via $p\,dp/du = d^2u/ds^2$ and integrating:

$$p^2 = E_{\text{total}} - \frac{L^2}{2(1+u)^2}.$$

From the metric normalisation $2p^2 + L^2/(1+u)^2 = 1$, the total energy is $E_{\text{total}} = 1/2 = g^{uu}$.

The position-dependent term is the geometric potential:

$$V(u) = \frac{k^2}{2(1+u)^2},$$

where we identify $L = k$ (the angular momentum equals the winding number).

Integration. The integrated potential per cell:

$$E_{\text{phase}} = \int_0^1 \frac{k^2}{2(1+u)^2}\,du.$$

By the layer-cake decomposition, the level width of $V(u) = k^2/(2(1+u)^2)$ (strictly decreasing on $[0,1]$) is:

$$w_V(t) = \begin{cases} 1 & 0 \le t < k^2/8, \\ \dfrac{k}{\sqrt{2t}} - 1 & k^2/8 \le t \le k^2/2. \end{cases}$$

Contraction integral:

$$\int_0^{k^2/2} w_V(t)\,dt = \frac{k^2}{8} + \frac{k^2}{8} = \frac{k^2}{4}.$$

$$E_{\text{phase}} = \frac{k^2}{4}.$$

Consistency check: Integrated kinetic energy + integrated potential $= (1/2 - k^2/4) + k^2/4 = 1/2 = g^{uu}$. ✓

Verified symbolically via SymPy.

Corollary 6 (Cell-Independence of the Local Metric)**.** ***Statement.*** *The quantities governing local physics on each cell are independent of the cell number n.*

Proof. Parameterise cell n with base radius a (where $a = n + 1$ for cell n in the global chain), so the embedding becomes:

$$\mathbf{R}_a(u, v) = \big((a + u)\cos\theta,\ (a + u)\sin\theta,\ u\big), \qquad \theta = v + 2\pi k u.$$

The metric components on this general cell are:

$$E(a) = 4\pi^2 k^2 (a + u)^2 + 2, \quad F(a) = 2\pi k\,(a + u)^2, \quad G(a) = (a + u)^2.$$

These do depend on a. However, the physically governing quantities do not:

1. *Gauge connection:*

$$A_u = \frac{F(a)}{G(a)} = \frac{2\pi k\,(a + u)^2}{(a + u)^2} = 2\pi k. \qquad \textit{(independent of } a\textit{)}$$

2. *Inverse metric:*

$$g^{uu} = \frac{G(a)}{\det g(a)} = \frac{(a + u)^2}{2(a + u)^2} = \frac{1}{2}. \qquad \textit{(independent of } a\textit{)}$$

3. *Gauge-diagonalised metric: The coordinate change $v' = v + 2\pi k u$ gives*

$$ds^2 = 2\,du^2 + (a + u)^2\,dv'^2$$

on every cell. The wave equation $\Box\psi = 0$ with $\psi = F(u)\,e^{i\ell v'}$ yields Euler's ODE:

$$(a + u)^2 F'' + (a + u)F' - 2\ell^2 F = 0,$$

whose indicial roots $r = \pm\ell\sqrt{2}$ are independent of a.

What does depend on a: The ratio $r(1)/r(0) = (a+1)/a$, which equals 2 only when $a = 1$. The transfer matrix trace $\operatorname{tr} T_\ell = \frac{3}{2}\cosh(\ell\sqrt{2}\ln 2)$ uses $\ln(w(1)/w(0)) = \ln 2$, which requires $a = 1$.

Conclusion. The chain is periodic: each cell resets to $a = 1$, with $u \in [0, 1]$ as a local coordinate. The gauge connection, the inverse metric, and the indicial structure of the wave equation are cell-independent. The transfer matrix (derived in later corollaries) requires this periodicity to achieve $\det T = 1/2$ and $\operatorname{tr} T_\ell = \frac{3}{2}\cosh(\ell\sqrt{2}\ln 2)$.

Verified symbolically via SymPy.

3. Part II: The Stacked Chain

Corollary 7 (Fourier Decomposition on the Transverse Circle)**.** *Axiom 6 requires $\psi(u, v + 2\pi) = \psi(u, v)$. A function $e^{i\ell v}$ satisfies this if and only if:*

$$e^{i\ell(v+2\pi)} = e^{i\ell v} \iff e^{2\pi i\ell} = 1 \iff \ell \in \mathbb{Z}.$$

Non-integer values fail: for example, $\ell = 1/2$ gives $e^{i\pi} = -1$, which would require $\psi(u, v + 2\pi) = -\psi(u, v)$—a weaker closure condition not permitted by Axiom 6.

The general solution on cell n is therefore:

$$\psi_n(u, v) = \sum_{\ell=-\infty}^{\infty} c_\ell^{(n)}\, f_\ell^{(n)}(u)\, e^{i\ell v}, \qquad \ell \in \mathbb{Z}.$$

No truncation. The full tower of integer angular momentum modes is present. The quantisation of ℓ is not imposed—it is forced by the closure condition.

Verified symbolically via SymPy (integer ℓ gives $e^{2\pi i\ell} = 1$; half-integer gives -1).

Corollary 8 (Per-Mode Radial ODE)**.** *Imported machinery. The Laplace–Beltrami operator is the unique second-order differential operator on a Riemannian manifold that is (a) coordinate-invariant, (b) self-adjoint, and (c) reduces to the ordinary Laplacian on flat space. It is not derived from the axioms. We apply known Riemannian geometry to the metric our axioms produced:*

$$\Box_M \psi = \frac{1}{\sqrt{g}} \frac{\partial}{\partial x^i} \left(\sqrt{g}\, g^{ij} \frac{\partial \psi}{\partial x^j} \right).$$

Two correction factors versus the flat Laplacian: g^{ij} accounts for non-orthogonal coordinate directions; $\sqrt{g}$ accounts for varying patch area across the surface.

In the original (u, v) frame. Substituting $\psi = f(u)\, e^{i\ell v}$ into $\Box_M \psi = 0$ and dividing by $e^{i\ell v}$ yields a second-order ODE for $f(u)$ that depends on both k (from the metric) and ℓ (from the Fourier mode). The full expression is large due to the off-diagonal metric coupling.

In the gauge-diagonalised frame. The coordinate change $v' = v + 2\pi k u$ reduces the metric to the cone $ds^2 = 2\, du^2 + (1+u)^2\, dv'^2$. The same Laplace–Beltrami operator with $\psi = F(u)\, e^{i\ell v'}$ yields Euler's ODE:

$$(1+u)^2\, F'' + (1+u)\, F' - 2\ell^2\, F = 0.$$

The indicial equation $r^2 = 2\ell^2$ gives roots $r = \pm\ell\sqrt{2}$, and the two linearly independent solutions are:

$$F_1(u) = (1+u)^{\ell\sqrt{2}}, \qquad F_2(u) = (1+u)^{-\ell\sqrt{2}}.$$

Verified symbolically via SymPy: both solutions give zero residual when substituted back into the ODE.

Corollary 9 (Phase-Matching Condition — Mode $\ell = k$)**.** *Each Fourier mode $e^{i\ell v}$ carries angular momentum ℓ. The gauge connection (Corollary 2) contributes holonomy $2\pi k$ per cell. The relative phase mismatch per cell is:*

$$\delta\varphi_\ell = 2\pi(\ell - k).$$

Resonant mode ($\ell = k$): $\delta\varphi_k = 0$. The mode propagates coherently across cells with no phase drift.

Non-resonant modes ($\ell \neq k$): Over N cells the accumulated phase is $2\pi N(\ell - k)$. For $N \gg 1$, these contributions destructively interfere and the mode decays.

This identifies $\ell = k$ as a natural phase-matching condition: the mode whose angular momentum matches the winding number is the one selected by coherence.

Caveat. This condition arises from phase alignment, not from a derived dynamical suppression law. No claim is made here that non-resonant modes are forbidden—only that they fail to propagate coherently across many cells. A rigorous decay estimate (e.g. bounding $\|T_\ell^N\|$ for $\ell \neq k$) would strengthen this to a derived selection rule, and is deferred to later work.

Verified symbolically via SymPy.

Corollary 10 (Structural Correspondences: Wave Equation, Factorisation, and Normal Form)**.** *The Euler ODE from Corollary 8,*

$$(1+u)^2\, F'' + (1+u)\, F' - 2\ell^2\, F = 0,$$

admits three algebraic reductions. Each exhibits structural correspondence with a known equation of physics. No ansatz is used—only operator algebra and change of dependent variable. The correspondences are presented as structural analogies, not claimed equivalences.

1. Wave equation structure (Klein–Gordon analogy). *The Euler ODE arises from $\Box_M \psi = 0$ after separation of variables. It is the full second-order wave equation on the manifold. This provides a Klein–Gordon-type structure, but relies on the imported Laplace–Beltrami operator (Corollary 8).*

2. Scalar first-order factorisation (Dirac analogy). *Define the Euler operator $D = (1+u)\, d/du$. Then:*

$$D^2 F = (1+u)\frac{d}{du}\big[(1+u)F'\big] = (1+u)^2 F'' + (1+u)F'.$$

So the Euler ODE is $D^2 F - 2\ell^2 F = 0$, which factors algebraically:

$$\big(D - \ell\sqrt{2}\big)\big(D + \ell\sqrt{2}\big)F = 0.$$

Each factor is a first-order equation:

$$(1+u)\, F' + \ell\sqrt{2}\, F = 0, \qquad (1+u)\, F' - \ell\sqrt{2}\, F = 0.$$

Each reproduces one of the two indicial solutions: $F = (1+u)^{-\ell\sqrt{2}}$ and $F = (1+u)^{+\ell\sqrt{2}}$ respectively.

Caveat. This is a scalar first-order factorisation. It is structurally analogous to Dirac's procedure of factoring a second-order operator into first-order pieces, but it does not introduce spinors, Clifford algebra, or additional internal degrees of freedom. The analogy is algebraic, not physical, until further structure is derived.

3. Schrödinger-type normal form. *To eliminate the first-derivative term from the Euler ODE algebraically, substitute $F = (1+u)^{\alpha}\, h(u)$ and choose α to kill the h' coefficient. The requirement is $2\alpha + 1 = 0$, giving $\alpha = -1/2$. With $F = (1+u)^{-1/2}\, h(u)$:*

$$(1+u)^2\, h'' + \big(\tfrac{1}{4} - 2\ell^2\big)\, h = 0.$$

No h' term remains. Dividing by $-2(1+u)^2$:

$$-\frac{1}{2}\, h'' + \frac{2\ell^2 - \frac{1}{4}}{2(1+u)^2}\, h = 0.$$

This is a Schrödinger-type normal form (Sturm–Liouville structure): $-\frac{1}{2}h'' + V_{\text{eff}}(u)\, h = 0$ with effective potential $V_{\text{eff}} = (2\ell^2 - 1/4)/(2(1+u)^2)$, which for $\ell \gg 1$ approaches the geodesic potential $\ell^2/(1+u)^2$ from Corollary 5.

Caveat. This is a time-independent normal form. It is not yet a full time-dependent Schrödinger equation; the role of u as an evolution parameter is not formally established within the present axioms.

4. Coordinate sensitivity: origin of i. *In the gauge-diagonalised cone frame, the ODE is purely real. In the original (u, v) frame, the inverse metric has $g^{uv} = -\pi k$, and the cross-term $g^{uv}\partial_u\partial_v\psi$ introduces imaginary contributions. On the mass shell ($\ell = k$), the coefficient of $f'(u)$ separates as:*

$$\text{coeff}(f') = \underbrace{\frac{1}{2(1+u)}}_{\textit{real: geometric damping}} - \underbrace{2i\pi k^2}_{\textit{imaginary: constant}}.$$

The imaginary part $-2i\pi k^2$ is constant because $g^{uv} = -\pi k$ is constant, and it vanishes in the cone frame because the gauge diagonalisation removes g^{uv} entirely.

Caveat. This imaginary structure is coordinate-dependent: it is present in the (u, v) frame and absent in the (u, v') frame. Its physical significance (if any) remains to be established. Both frames describe the same geometry; the i is a feature of the chart, not yet shown to be a feature of the physics.

Summary. *Three levels of the same equation:*

Structure	*Operation*	*Result*	*Status*
Wave eqn	*Full Euler ODE*	$(1+u)^2F'' + (1+u)F' - 2\ell^2 F = 0$	*Imported* $\Box_M$
Factorisation	$(D-r)(D+r)F = 0$	*Two first-order equations*	*Scalar only*
Normal form	$F = (1+u)^{-1/2}h$	$-\frac{1}{2}h'' + V_{\text{eff}}\, h = 0$	*Time-indep.*

Verified symbolically via SymPy: operator factorisation residuals zero; normal-form reduction confirmed; f' coefficient splits into real and imaginary parts as stated.

4. Interpretation Boundary

The present work distinguishes between three categories of result:

Derived from axioms and standard differential geometry: Metric · Connection · Holonomy · Christoffel symbols · Geodesics · Gaussian curvature · Cell action · Cell-independence · Fourier quantisation · Phase-matching condition

Imported from standard Riemannian geometry: Laplace–Beltrami operator · Wave equation $\Box_M \psi = 0$

Interpretive (structural correspondence, not equivalence): Klein–Gordon analogy · Dirac-like factorisation · Schrödinger normal form · Phase-matching as "mass shell" · Imaginary coefficient as physical i

These correspondences are suggestive. Whether any of them can be promoted to physical equivalences requires: (a) a derived identification of u with a physical evolution parameter; (b) an observer or measurement framework; and (c) predictions that can be tested against experiment.

Part III: Inter-Cell Matching and Spectrum

Corollary 11 (Inter-Cell Transfer Matrix)**.** *At cell junctions, the wave equation $\Box_M \psi = 0$ requires continuity of both the mode function and its derivative:*

$$f_\ell^{(n)}(1) = f_\ell^{(n+1)}(0), \qquad \left.\frac{df_\ell^{(n)}}{du}\right|_{u=1} = \left.\frac{df_\ell^{(n+1)}}{du}\right|_{u=0}.$$

Why C^1. The ODE is second-order. A discontinuity in f' at a junction would produce a delta function in f'', requiring a point source at the boundary. Since no sources are postulated, both f and f' must be continuous. Two matching conditions at each junction, two linearly independent solutions per cell—the propagation is completely determined by a 2×2 matrix.

Construction. From Corollary 8, the gauge-diagonalised cone metric yields Euler's equation with solutions $F_1 = (1+u)^p$ and $F_2 = (1+u)^{-p}$, where $p = \ell\sqrt{2}$. Their values and derivatives at the cell boundaries are:

$$F_1(0) = 1, \qquad F_1'(0) = p, \qquad F_1(1) = 2^p, \qquad F_1'(1) = p \cdot 2^{p-1},$$
$$F_2(0) = 1, \qquad F_2'(0) = -p, \qquad F_2(1) = 2^{-p}, \qquad F_2'(1) = -p \cdot 2^{-p-1}.$$

The transfer matrix is built by propagating two canonical initial conditions:

Column 1 ($f(0) = 1$, $f'(0) = 0$): $A = B = 1/2$.

$$f(1) = \frac{2^p + 2^{-p}}{2} = \cosh(p \ln 2), \qquad f'(1) = \frac{p}{2} \sinh(p \ln 2).$$

Column 2 ($f(0) = 0$, $f'(0) = 1$): $A = 1/(2p)$, $B = -1/(2p)$.

$$f(1) = \frac{\sinh(p \ln 2)}{p}, \qquad f'(1) = \frac{1}{2} \cosh(p \ln 2).$$

Therefore:

$$T_\ell = \begin{pmatrix} \cosh(p \ln 2) & \dfrac{\sinh(p \ln 2)}{p} \\ \dfrac{p}{2} \sinh(p \ln 2) & \dfrac{1}{2} \cosh(p \ln 2) \end{pmatrix}, \qquad p = \ell\sqrt{2}.$$

Immediate properties.

$$\operatorname{tr} T_\ell = \cosh(p \ln 2) + \tfrac{1}{2} \cosh(p \ln 2) = \tfrac{3}{2} \cosh(\ell\sqrt{2} \ln 2).$$

$$\det T_\ell = \tfrac{1}{2} \cosh^2(p \ln 2) - \tfrac{1}{2} \sinh^2(p \ln 2) = \tfrac{1}{2}.$$

The determinant is not unity—it equals $\sqrt{g(0)}/\sqrt{g(1)} = \sqrt{2}/(2\sqrt{2}) = 1/2$, recording the metric expansion from cell entry to cell exit (Corollary 13 derives this from the Wronskian).

Degenerate case ($\ell = 0$): The double root $p = 0$ gives $F_1 = 1$, $F_2 = \ln(1 + u)$, and:

$$T_0 = \begin{pmatrix} 1 & \ln 2 \\ 0 & 1/2 \end{pmatrix}.$$

Propagation. By Corollary 6, each cell has the same local metric, hence the same T_ℓ. After N cells:

$$\begin{pmatrix} f(N) \\ f'(N) \end{pmatrix} = T_\ell^N \begin{pmatrix} f(0) \\ f'(0) \end{pmatrix}.$$

Riccati perspective. The substitution $z = F'/F$ converts the Euler ODE to the Riccati equation:

$$\frac{dz}{du} = -z^2 - \frac{z}{1+u} + \frac{2\ell^2}{(1+u)^2}.$$

Under the log coordinate $s = \ln(1 + u)$ and the rescaling $Z = (1 + u)\, z$, this autonomises to:

$$\frac{dZ}{ds} = -Z^2 + 2\ell^2,$$

with constant fixed points $Z_\pm = \pm\ell\sqrt{2}$. Both fixed points live in the same coordinate—there is no regime transition, no coordinate incompatibility, no special function. This is the trivial case of Riccati flow: WKB is exact on the cone because the Euler equation has constant coefficients in log coordinates.

The two columns of T_ℓ correspond to the two fixed-point trajectories: each solution $F = (1 + u)^{\pm p}$ follows a single Riccati fixed point from $u = 0$ to $u = 1$ without deviation.

Connection to the special functions framework. The companion paper "What Is So Special About 'Special Functions'?" (Yazdani, 2026) argues that every classical special function is a smooth Riccati flow trajectory connecting two asymptotic regimes with incompatible autonomising coordinates. The Euler equation on the cone is the degenerate case: only one autonomising coordinate ($\ln w$) is needed, both regimes are compatible, and the flow reduces

to constant fixed points. No scar forms. The transfer matrix T_ℓ is the finite propagator of this trivial flow.

Verification.

- *Symbolic: Residuals of both solutions $F = (1+u)^{\pm\ell\sqrt{2}}$ vanish identically when substituted into the Euler ODE (SymPy).*
- *Numeric spot checks: Residuals exactly 0 across $\ell \in \{1,2,3,5,50\}$, $u \in \{1,2,3,5,50\}$ (SymPy, exact rational arithmetic).*
- *Riccati solver: Autonomous classification recovered; both fixed points identified; no poles on $[0,1]$.*
- *Numerical integration: Solution values, transfer matrix entries, and Riccati flow confirmed to machine precision (scipy, errors $< 10^{-11}$).*
- *Transfer matrix: $\det T_\ell = 1/2$ and $\operatorname{tr} T_\ell = \frac{3}{2}\cosh(\ell\sqrt{2}\ln 2)$ confirmed for $\ell = 0,1,2,3,5$ (scipy, errors $< 10^{-10}$).*

Corollary 12 (Holonomy Coherence)**.** *To be promoted to revised standard. Sketch: Corollary 9 establishes the phase-matching condition $\ell = k$. This corollary formalises the coherence argument: on the mass shell, $\delta\varphi_k = 0$ and modes propagate without phase drift; off-shell, accumulated phase $2\pi N(\ell - k)$ produces destructive interference for $N \gg 1$.*

Corollary 13 (Wronskian Conservation and $\det T = \frac{1}{2}$)**.** *To be promoted to revised standard. Sketch: The Wronskian $W(u) = \sqrt{g(u)}\,(f_1 f_2' - f_2 f_1')$ satisfies $dW/du = 0$. Since $\sqrt{g(0)} = \sqrt{2}$ and $\sqrt{g(1)} = 2\sqrt{2}$, the transfer matrix determinant is $\det T_\ell = \sqrt{g(0)}/\sqrt{g(1)} = 1/2$.*

Corollary 14 (Dispersion Relation)**.** *To be promoted to revised standard. Sketch: Eigenvalues of T_ℓ are $\lambda_\pm = \frac{1}{\sqrt{2}}e^{\pm i\kappa}$. Bloch phase: $\cos\kappa = \frac{1}{\sqrt{2}}\operatorname{tr} T_\ell$. Propagating modes: $|\operatorname{tr} T_\ell| \le \sqrt{2}$. Evanescent modes: $|\operatorname{tr} T_\ell| > \sqrt{2}$.*

Corollary 15 (Chain Phase, Action, and Energy)**.** *To be promoted to revised standard. Sketch: Over an N-cell chain, $\Phi_{\text{total}} = 2\pi k N$, $S_{\text{total}} = 3\sqrt{2}\,\pi N$, $E_{\text{total}} = (k^2/4)\,N$. All extensive in N.*

Part IV: Intrinsic Flatness and Field Equations

Corollary 16 (Gaussian Curvature)**.** *To be promoted to revised standard. Sketch: $K(u) = 0$ for all $u \in [0,1]$ and all k. The manifold is intrinsically flat—a developable surface. All curvature experienced by modes is extrinsic.*

Corollary 17 (Extrinsic Curvature)**.** *To be promoted to revised standard. Sketch: Second fundamental form coefficients L, M, N computed from the unit normal. $LN - M^2 = 0$ confirms $K = 0$ (rank-1, developable). Mean curvature $H = \sqrt{2}/(4(1+u))$.*

Corollary 18 (Gauss–Bonnet and Topology)**.** *To be promoted to revised standard. Sketch: $K = 0$ identically $\Rightarrow \int K\,dA = 0$. Each cell is an annulus ($\chi = 0$). The topology is trivial.*

Corollary 19 (Maxwell's Equations from Gauge Coupling)**.** ***Statement.*** *In the gauge-diagonalised cone frame, the wave equation $\Box_M \psi = 0$ is purely real: the Euler ODE for $F(u)$ has no coupling between real and imaginary parts. In the original (u,v) frame, the off-diagonal inverse metric $g^{uv} = -\pi k$ reintroduces the gauge connection, and the radial ODE acquires complex coefficients. Separating $f = a + ib$ (where $a = \operatorname{Re}(f)$, $b = \operatorname{Im}(f)$)*

yields a pair of coupled real equations whose cross-coupling has the antisymmetric structure of Maxwell's equations.

Derivation. The Laplace–Beltrami operator in the original (u, v) frame, applied to $\psi = f(u)\, e^{i\ell v}$ and divided by $e^{i\ell v}$, gives:

$$\frac{1}{\sqrt{g}}\partial_i\big(\sqrt{g}\, g^{ij}\, \partial_j\psi\big) \longrightarrow C_2\, f'' + C_1\, f' + C_0\, f = 0,$$

where the coefficients are computed from the four flux contributions ($g^{uu}\partial_u\partial_u$, $g^{uv}\partial_u\partial_v$, $g^{vu}\partial_v\partial_u$, $g^{vv}\partial_v\partial_v$) with each ∂_v replaced by $i\ell$. After clearing the denominator $(1+u)^2$:

$$C_2 = \tfrac{1}{2}(1+u)^2, \tag{7}$$

$$C_1 = \tfrac{1}{2}(1+u) - 2i\pi\ell k\,(1+u)^2, \tag{8}$$

$$C_0 = \ell\big(-2\pi^2\ell k^2(1+u)^2 - \ell - i\pi k(1+u)\big). \tag{9}$$

On the mass shell $\ell = k$ (Corollary 9), these become:

$$C_2 = \tfrac{1}{2}(1+u)^2, \qquad \textit{(real)}$$

$$C_1 = \underbrace{\tfrac{1}{2}(1+u)}_{\textit{real: damping}} - \underbrace{2i\pi k^2(1+u)^2}_{\textit{imaginary: curl coupling}}, \qquad \textit{(complex)}$$

$$C_0 = \underbrace{-2\pi^2k^4(1+u)^2 - k^2}_{\textit{real: potential}} - \underbrace{i\pi k^2(1+u)}_{\textit{imaginary: source coupling}}. \qquad \textit{(complex)}$$

Dividing through by $C_2 = \frac{1}{2}(1+u)^2$ gives the normalised ODE:

$$f'' + \left(\frac{1}{1+u} - 4i\pi k^2\right) f' + \left(-V - \frac{2i\pi k^2}{1+u}\right) f = 0,$$

where $V = (4\pi^2k^4(1+u)^2 + 2k^2)/(1+u)^2$ is the effective potential (real).

Define:

$$\gamma = 4\pi k^2 \quad \textit{(constant)}, \qquad \sigma = \frac{2\pi k^2}{1+u} \quad \textit{(position-dependent)}.$$

The coupled real system. Substituting $f = a(u) + i\, b(u)$ and separating real and imaginary parts, the single complex ODE splits into:

$$\textit{Real equation:} \quad \mathcal{L}[a] = -\gamma\, b' - \sigma\, b,$$

$$\textit{Imaginary equation:} \quad \mathcal{L}[b] = +\gamma\, a' + \sigma\, a,$$

where $\mathcal{L}[\,\cdot\,] = (\cdot)'' + \frac{1}{1+u}(\cdot)' - V(\cdot)$ is the scalar wave operator, identical for both components.

The cross-coupling terms are:

Coupling	*In $\mathcal{L}[a]$ eqn*	*In $\mathcal{L}[b]$ eqn*	*Structure*
First-order (curl)	$-\gamma\, b' = -4\pi k^2 b'$	$+\gamma\, a' = +4\pi k^2 a'$	*Antisymmetric, constant*
Zeroth-order (source)	$-\sigma\, b = -\frac{2\pi k^2}{1+u} b$	$+\sigma\, a = +\frac{2\pi k^2}{1+u} a$	*Antisymmetric, decays*

Both couplings are antisymmetric: the coefficient of b' in the real equation is the negative of the coefficient of a' in the imaginary equation, and likewise for b and a. This is the defining algebraic signature of electromagnetism.

Recovery of standard Maxwell. In the flat limit (cone corrections removed: $1/(1+u) \to 0$, $V \to$ const, $\sigma \to 0$), the coupled system reduces to:

$$a'' = -\gamma\, b', \qquad b'' = +\gamma\, a'.$$

Integrating once in u (constant of integration zero for wave solutions):

$$\frac{da}{du} = -\gamma\, b = -4\pi k^2\, b, \tag{10}$$

$$\frac{db}{du} = +\gamma\, a = +4\pi k^2\, a. \tag{11}$$

Under the identification $a \leftrightarrow E$ (electric field), $b \leftrightarrow B$ (magnetic field), $u \leftrightarrow z$ (spatial coordinate), these are Maxwell's curl equations in vacuum for a monochromatic plane wave propagating in the z-direction:

$$\frac{dE}{dz} = -\omega\, B, \qquad \frac{dB}{dz} = +\omega\, E, \qquad \omega = 4\pi k^2.$$

Differentiating the first and substituting the second gives the wave equation:

$$\frac{d^2E}{dz^2} + \omega^2\, E = 0.$$

Phase velocity: $v = \omega/\kappa = 1 = c$, by Axiom 1.

The Riemann–Silberstein vector. The complex field $f = a + ib$ is the Riemann–Silberstein vector $\mathbf{F} = E + iB$, introduced by Silberstein in 1907 as a reformulation of Maxwell's equations into a single complex equation. The present derivation inverts this: the helix produces f as a single object; E and B are its real and imaginary projections. The Riemann–Silberstein vector is not a mathematical convenience—it is the natural field.

Geometric origin. The cross-coupling exists because $g^{uv} = -\pi k \neq 0$ in the original frame. In the gauge-diagonalised frame ($v' = v + 2\pi k u$), $g^{uv} = 0$ and the ODE is purely real—no coupling, no electromagnetism. The gauge transformation that removes A_u simultaneously removes the electromagnetic structure.

The three corrections that distinguish Maxwell-on-the-cone from flat-space Maxwell are:

1. *Geometric damping: $P = 1/(1+u)$. Both E and B have a first-derivative self-term encoding $1/r$ amplitude decay from the expanding cone.*
2. *Effective potential: $V(u)$. Second-order self-coupling containing the centrifugal term $k^2/(1+u)^2$ and the gauge-squared term $4\pi^2 k^4$.*
3. *Geometric source: $\sigma = 2\pi k^2/(1+u)$. Position-dependent zeroth-order cross-coupling absent in flat space, encoding how the expanding geometry creates an effective charge density. Antisymmetric: no net monopole.*

What $\gamma = 4\pi k^2$ means. The coupling constant is the winding number squared. It is k^2 because on the mass shell $\ell = k$: one factor of k comes from the metric (Axiom 4) and one from the Fourier mode (Corollary 7). Electromagnetic charge is topology.

Cone frame interpretation.

- *Gauge-diagonalised frame: $g^{uv} = 0$. Purely real ODE. No coupling between a and b. Pure geometry. No electromagnetism.*
- *Original frame: $g^{uv} = -\pi k$. Complex ODE. a and b couple antisymmetrically. Maxwell's equations emerge.*

The "two polarisations of light" are not fundamental. Light is one helical object $f(u)\, e^{i\ell v}$ propagating on the manifold. Electromagnetism is the projection of helical geometry onto a real basis.

Caveat. This derivation recovers the structural form of Maxwell's equations from the wave equation on the helix. The identification $a \leftrightarrow E$, $b \leftrightarrow B$ is a structural correspondence: it

matches the algebraic signature (antisymmetric first-order cross-coupling) and recovers the correct flat-limit equations. Promoting this to a physical equivalence requires: (a) identifying the electromagnetic field strength $F_{\mu\nu}$ with a geometric quantity on the manifold; (b) recovering the Lorentz force law from geodesic deviation; and (c) matching the coupling constant $4\pi k^2$ to the observed electromagnetic coupling $e^2/(4\pi\epsilon_0)$. These are deferred to later work.

Verification. All coefficients C_2, C_1, C_0 computed symbolically via the Laplace–Beltrami operator in SymPy. Real/imaginary splitting confirmed. Antisymmetry of cross-coupling verified algebraically. Cone frame limit ($g^{uv} \to 0$) confirmed to annihilate all imaginary coefficients. Flat-limit Maxwell equations recovered and verified to produce the wave equation with phase velocity $c = 1$. Full computation in `maxwell_helix_full.py`.

Corollary 20 (Gravity from Chain Curvature)**.** ***Statement.*** *The intrinsically flat helical manifold ($K = 0$) carries no gravitational field at the single-cell level. Gravity emerges on the* coarse-grained *manifold obtained by filling in the inter-winding gaps of the stacked chain. The effective 4-metric on this manifold inherits a non-vanishing torsion from the helical winding structure; the symmetric part of the resulting connection reproduces gravitational phenomenology. The geodesic equations on this manifold reduce to a Riccati equation whose flow rate is controlled by $\alpha'(R)$—the derivative of the running fine structure constant—identifying gravity as a second-order geometric effect.*

Coarse-grained coordinates. *The stacked chain is embedded in a torus of major radius R (cosmological, expanding) and minor radius r (transverse). At the single-cell level, the coordinates are (u, v) on the flat sheet. At the coarse-grained level, two additional coordinates appear:*

- *w: the inter-winding direction, perpendicular to the sheet within the torus body. Because adjacent windings are offset in v, the w-direction is not purely radial but carries a helical twist—this is the torsion of the coarse-grained connection.*
- *R: the expanding torus radius. This is the cosmological coordinate, already axiomatic (Axiom 1 fixes $c = 1$; the torus expands monotonically).*

The signature of R is timelike: the transfer matrix has $\det(T) = 1/2$ per cell (exponential contraction of phase space volume along the chain), while R is expanding. These opposing monotonic flows create the signature split $(-, +, +, +)$.

Pitch angle from closure. *On the developed rectangle (width $2\pi r$, height $2\pi R$), the helix is a straight line. A (p, q) helix winds p times around the minor circle for every q trips around the major circle. Closure demands $p, q \in \mathbb{Z}$. The pitch angle satisfies*

$$\tan\theta = \frac{p\,r}{q\,R}.$$

The geometric ratio $1/(14\pi^2) \approx \alpha$ (established in earlier work) identifies the pitch angle with the fine structure constant. For the simplest closure $(p, q) = (1, 1)$:

$$\frac{R}{r} = 14\pi^2 \approx 138.2,$$

placing the torus aspect ratio within the toroidal correction of $1/\alpha_{\text{CODATA}} \approx 137.036$. As R expands, the aspect ratio shifts and the pitch angle runs: $\alpha = \alpha(R)$.

The effective metric. *Each piece is derived, not postulated:*

- *$du^2 + dv^2$: flat sheet ($K = 0$, Corollary 16).*

- *dw^2: coarse-grained inter-winding direction, constructed from the chain.*
- *$2\alpha(R)\,dv\,dw$: off-diagonal term encoding the helical twist (pitch angle from closure condition).*
- *$-f(R)\,dR^2$: timelike direction; $f(R)$ determined by $\det(T) = 1/2$ scaling with the expanding radius.*

The line element is:

$$ds^2 = -f(R)\,dR^2 + du^2 + dv^2 + dw^2 + 2\alpha(R)\,dv\,dw.$$

Determinant: $\det(g) = -f(R)(1-\alpha(R)^2)$. *Inverse metric:*

$$g^{RR} = -\frac{1}{f}, \quad g^{uu} = 1, \quad g^{vv} = \frac{-1}{\alpha^2-1}, \quad g^{vw} = \frac{\alpha}{\alpha^2-1}, \quad g^{ww} = \frac{-1}{\alpha^2-1}.$$

Christoffel symbols. *Six independent non-zero components, all involving $f'(R)$ or $\alpha'(R)$:*

$$\Gamma^R_{RR} = \frac{f'}{2f}, \qquad \Gamma^R_{vw} = \frac{\alpha'}{2f}, \tag{12}$$

$$\Gamma^v_{Rv} = \frac{\alpha\,\alpha'}{2(\alpha^2-1)}, \qquad \Gamma^v_{Rw} = \frac{-\alpha'}{2(\alpha^2-1)}, \tag{13}$$

$$\Gamma^w_{Rv} = \frac{-\alpha'}{2(\alpha^2-1)}, \qquad \Gamma^w_{Rw} = \frac{\alpha\,\alpha'}{2(\alpha^2-1)}. \tag{14}$$

The u-direction is geodesically free: all Christoffel symbols with a u-index vanish. Crucially, $\Gamma^R_{vw} = \alpha'(R)/(2f(R))$ couples motion in v and w to acceleration in R. This is the gravitational term.

Geodesic equations. *The four geodesic equations $\ddot{x}^\sigma + \Gamma^\sigma_{\mu\nu}\,\dot{x}^\mu\dot{x}^\nu = 0$ are:*

$$\ddot{R} + \frac{f'}{2f}\,\dot{R}^2 + \frac{\alpha'}{f}\,\dot{v}\,\dot{w} = 0, \tag{15}$$

$$\ddot{u} = 0, \tag{16}$$

$$\ddot{v} + \frac{\alpha'(\alpha\,\dot{v} - \dot{w})}{\alpha^2-1}\,\dot{R} = 0, \tag{17}$$

$$\ddot{w} + \frac{\alpha'(\alpha\,\dot{w} - \dot{v})}{\alpha^2-1}\,\dot{R} = 0. \tag{18}$$

*Equation (**??**) is free: u decouples completely. Equation (**??**) contains the gravitational coupling: motion in v and w produces acceleration in R, weighted by $\alpha'(R)/f(R)$.*

Riccati reduction. *At leading order ($\alpha^2 \ll 1$), the v–w system reparametrised from affine parameter s to R gives:*

$$\frac{dV}{dR} = -\alpha'(R)\,W, \qquad \frac{dW}{dR} = -\alpha'(R)\,V,$$

where $V = dv/ds$, $W = dw/ds$. The velocity ratio $y(R) = V/W$ satisfies:

$$\boxed{\frac{dy}{dR} = \alpha'(R)\,(y^2-1).}$$

This is a Riccati equation with coefficients $a = -\alpha'$, $b = 0$, $c = \alpha'$.

Fixed points and exact solution. *The fixed points are $y = \pm 1$ ($V = \pm W$: null trajectories on the v–w plane). The equation is separable; integrating:*

$$y(R) = -\coth\big(\alpha(R) + C\big),$$

where C is the integration constant set by initial conditions.

Stability analysis. Near $y = +1$: perturbations grow as $\exp(+2\alpha(R))$ (unstable). Near $y = -1$: perturbations decay as $\exp(-2\alpha(R))$ (stable). Since $\alpha \approx 1/137$, the flow rate is $\sim 2/137$ per R-unit. This is why gravitationally bound systems are quasi-stable over cosmological timescales.

Riemann and Ricci tensors. *The full Riemann tensor has 12 independent non-zero components, all built from α', α'', f', and f. The non-zero Ricci components are:*

$$R_{RR} = \frac{-\big(2(\alpha\,\alpha'' + \alpha'^2)(\alpha^2 - 1)f - (\alpha^2 - 1)\alpha\,\alpha' f' - 3\alpha^2 f\,\alpha'^2 + f\,\alpha'^2\big)}{2(\alpha^2 - 1)^2 f}, \tag{19}$$

$$R_{vv} = R_{ww} = \frac{\alpha'^2}{2(\alpha^2 - 1)f}, \tag{20}$$

$$R_{vw} = \frac{2f\,\alpha'' - \alpha' f'}{4f^2}. \tag{21}$$

The equality $R_{vv} = R_{ww}$ reflects isotropy in the v–w plane. The Ricci tensor vanishes identically when $\alpha'(R) = 0$ (constant coupling $\Rightarrow$ flat space $\Rightarrow$ no gravity).

Einstein tensor. *$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} \mathcal{R}$, where $\mathcal{R}$ is the Ricci scalar. Key structural properties:*

1. *$G_{vv} = G_{ww}$: isotropic pressure in the v–w plane.*
2. *$G_{uu} \neq G_{vv}$: the helix picks a preferred axis. The effective stress-energy is anisotropic—one direction along the helix, isotropic transverse to it.*
3. *All components vanish when $\alpha'(R) = 0$.*

Recovery of GR: Newtonian limit. *In the slow-field approximation (drop the $\dot{R}^2$ term in (**??**)):*

$$\ddot{R} \approx -\frac{\alpha'(R)}{f(R)}\,\dot{v}\,\dot{w}.$$

Comparing with Newton's law $\ddot{r} = -d\Phi/dr$:

$$\frac{d\Phi}{dr} \sim \frac{\alpha'(R)}{f(R)}\,\dot{v}\,\dot{w}.$$

The product $\dot{v}\,\dot{w}$ plays the role of the source (mass/energy). The ratio $\alpha'(R)/f(R)$ is the gravitational coupling.

Recovery of GR: Einstein equation as tautology. *In GR, $G_{\mu\nu} = 8\pi G\,T_{\mu\nu}$ relates geometry (left) to matter (right). In GToR, $G_{\mu\nu}$ is computed entirely from $f(R)$ and $\alpha(R)$, both derived from the helix geometry. There is no independent $T_{\mu\nu}$. What GR calls "matter" is the non-trivial part of the Einstein tensor. The Einstein equation is not wrong—it is a* tautology *when viewed from the helix: both sides are the same geometric object, split artificially into "geometry" and "matter" by the GR formalism.*

The hierarchy problem. *At leading order,* $\alpha = 1/(14\pi^2) = const$*, so* $\alpha'(R) = 0$ *and there is no gravity. Gravity requires the toroidal correction* $\varepsilon(R)$ *to the fine structure constant:*

$$\alpha(R) = \frac{1}{14\pi^2} + \varepsilon(R), \qquad \textit{gravitational coupling} = \frac{\varepsilon'(R)}{f(R)}.$$

Electromagnetism is the in-sheet coupling (Corollary 19), entering at $O(\alpha)$*. Gravity is the inter-sheet coupling, entering at* $O(\alpha')$*—the derivative of a* $1/137$*-scale correction. The hierarchy* $G_{gravity} \ll G_{EM}$ *is not a mystery; it is the difference between a function and its derivative.*

Connection to the Riccati framework. *The geodesic Riccati equation* $dy/dR = \alpha'(R)(y^2 - 1)$ *is the same structural form as the Riccati equations governing radial dynamics on the single cell (Corollary 11) and special function classification (Yazdani, 2026b). In all three cases:*

- *Fixed points are the asymptotic regimes.*
- *The flow rate is controlled by a geometric parameter.*
- *The solution is a hyperbolic function of the running variable.*

The Euler equation on the cone (Corollary 8) is the trivial *Riccati case: both fixed points live in the same log coordinate, WKB is exact, no scar forms. The geodesic equation on the coarse-grained manifold is the next case: the flow is slow (*$\sim 2\alpha$*) because the pitch angle is small. This is why gravitationally bound orbits persist over cosmological timescales.*

Caveat. *This corollary derives the effective metric, Christoffel symbols, geodesic equations, Riemann tensor, Ricci tensor, Einstein tensor, Newtonian limit, and Riccati reduction from the coarse-grained chain geometry. The following remain open:*

(a) *The explicit form of* $f(R)$*, to be derived from* $\det(T) = 1/2$ *scaling with the expanding torus radius.*
(b) *The explicit form of* $\varepsilon(R)$ *(the toroidal correction to* α*), to be derived from the closure condition at finite* R*.*
(c) *Recovery of the* $1/r^2$ *force law from the geodesic equation in the appropriate limit.*
(d) *Connection between the product* $\dot{v}\,\dot{w}$ *and the mass* $m = \pi k^2/2$ *derived from holonomy.*

These are deferred to later work.

Verification. *All computations performed symbolically via SymPy.*

- *Metric, inverse metric, and* $g \cdot g^{-1} = I$ *confirmed.*
- *All six independent Christoffel symbols computed; lower-index symmetry verified.*
- *Four geodesic equations derived from Christoffel symbols.*
- *Riccati reduction:* $y = -\coth(\alpha(R) + C)$ *substituted into* $dy/dR = \alpha'(y^2 - 1)$*; residual zero.*
- *Riemann tensor: 12 independent non-zero components computed.*
- *Ricci tensor:* $R_{vv} = R_{ww}$ *confirmed (v–w isotropy).*
- *Einstein tensor:* $G_{vv} = G_{ww}$ *confirmed.*
- *All tensors vanish identically when* $\alpha' = 0$*.*

Full computation in `gtor_cor20_verify.py`*.*